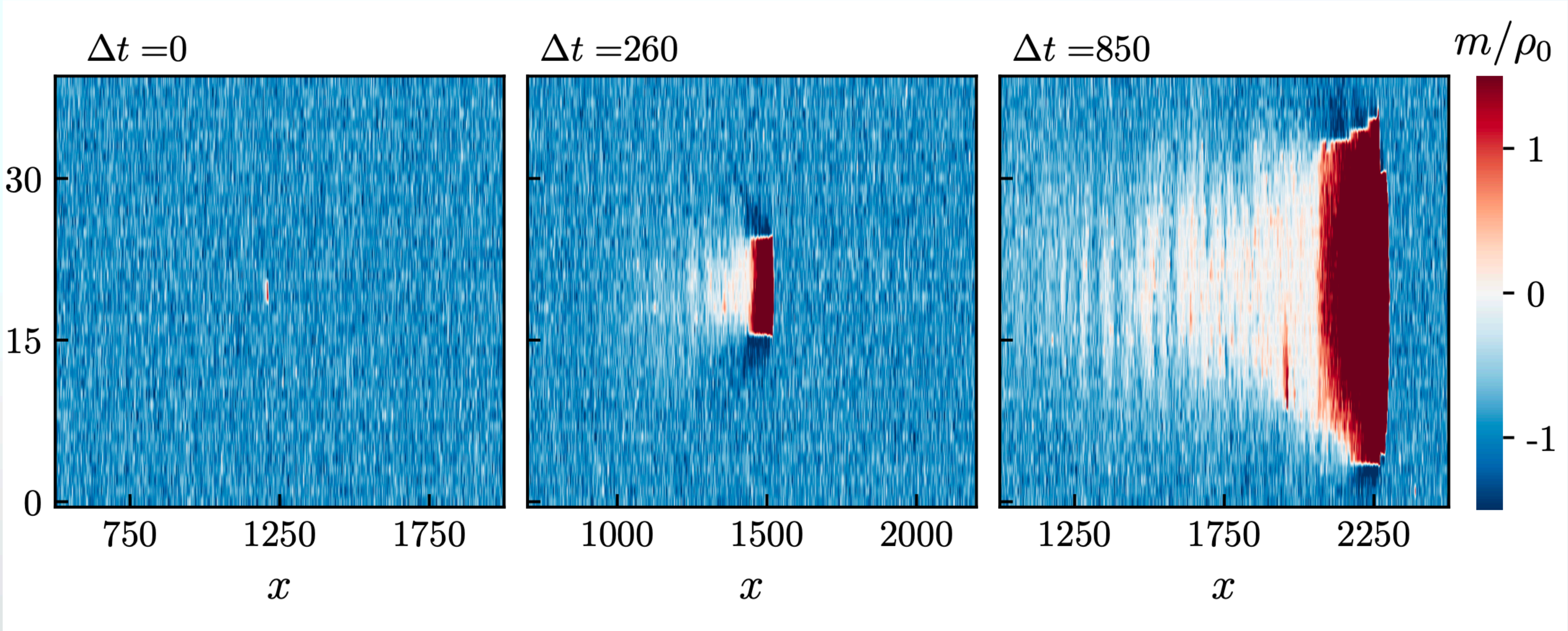
Metastability of Discrete-Symmetry Flocks

Brieuc Benvegnen et al. (2023)



Daniel Haim Breger, January 28th 2026

Background

- Equilibrium ordered phase is expected to be stable in $d > 1$ (Mermin-Wagner).
  - Finite perturbations expected to shrink
- Recent results questioned this:
  - Small obstacle can break the order in noisy system
  - Constant-density flocks are metastable to spontaneous nucleation

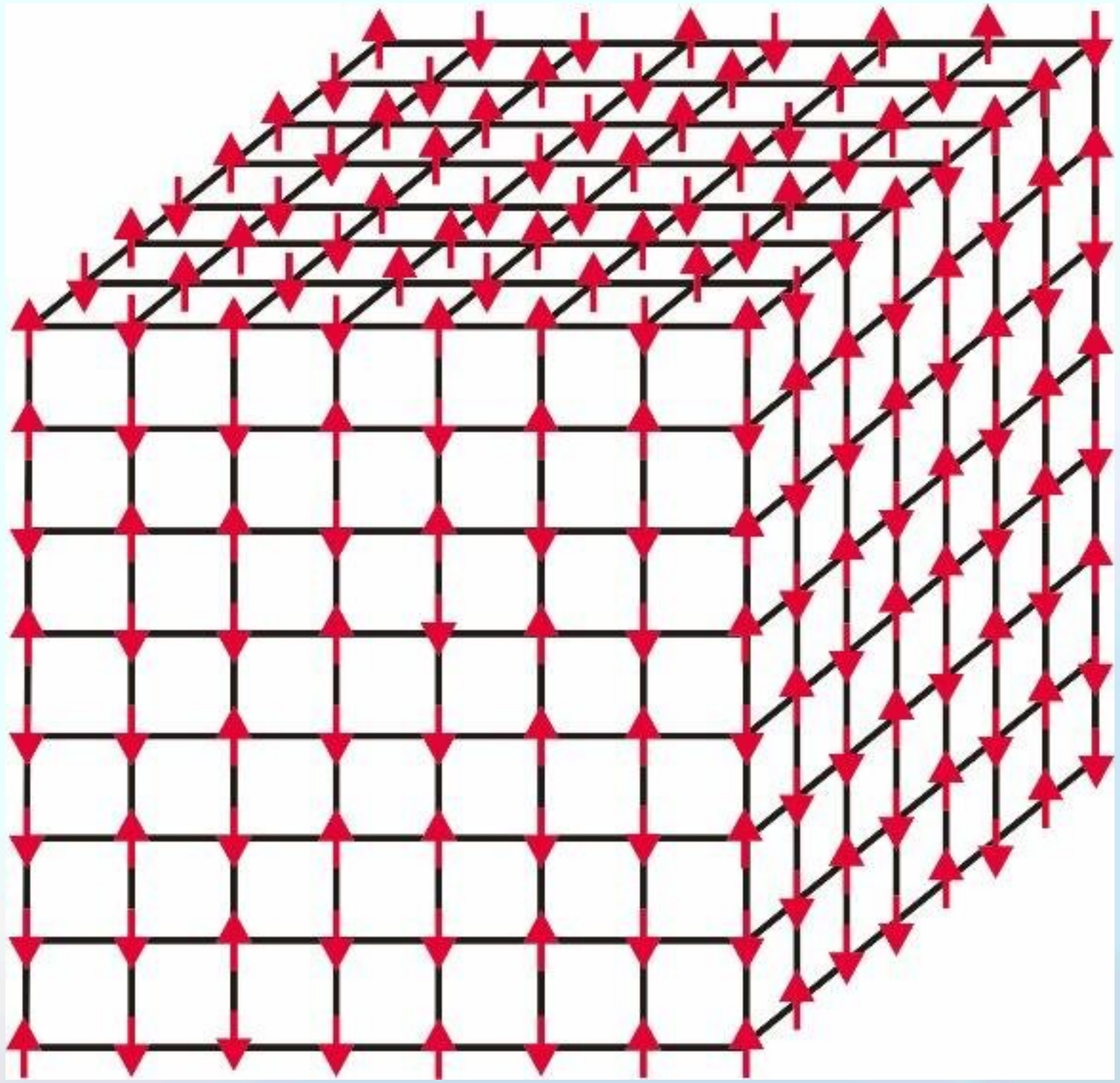
In This Paper

- **Can a flocking transition be induced for any system size? Yes.**
- **Can this transition happen spontaneously? Yes.**
- Flocking scalar polar models of $d \geq 2$ shown to have metastable ordered phase
  - Finite droplet of opposite polarization grows ballistically, taking over system
- Shown numerically and analytically

The Model

Active Ising Model

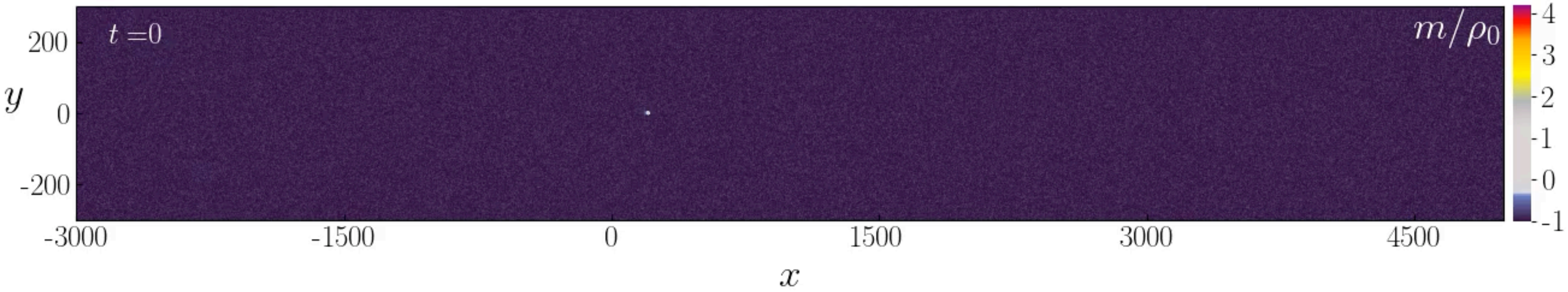
- spin: $s = \pm 1$
- Position: $\vec{i}_{n+1} = \vec{i}_n + s\hat{x}$ plus diffusion with rate D
- Occupancy at each site unrestricted
- On-site ferromagnetic spin flipping
- $\rho_{\vec{i}} = n^+_{\vec{i}} + n^-_{\vec{i}} \quad \& \quad m_{\vec{i}} = n^+_{\vec{i}} - n^-_{\vec{i}}$

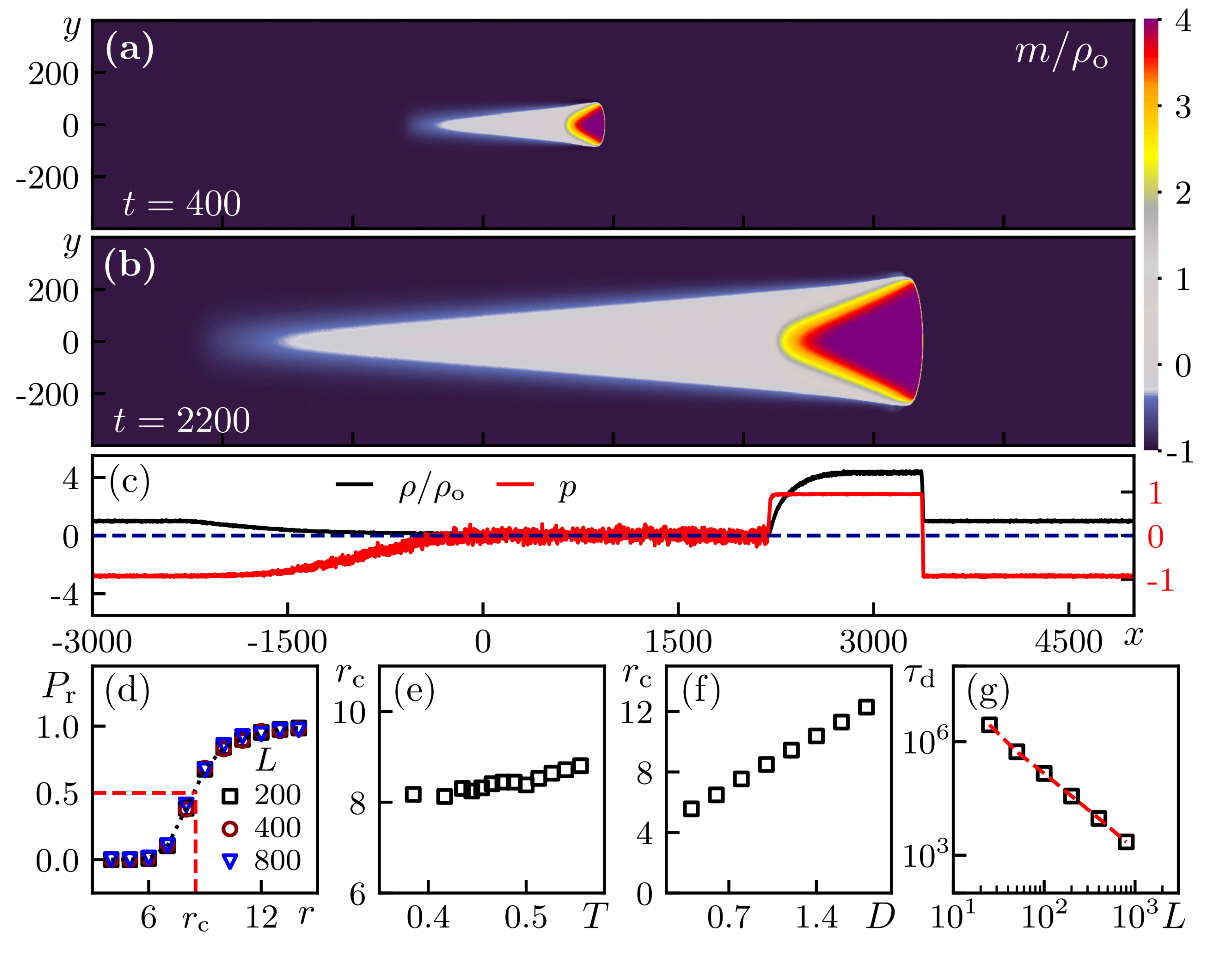


3D Ising Model Illustration. Zhang, Zhidong. (2025)

Induced Transition

In 2D

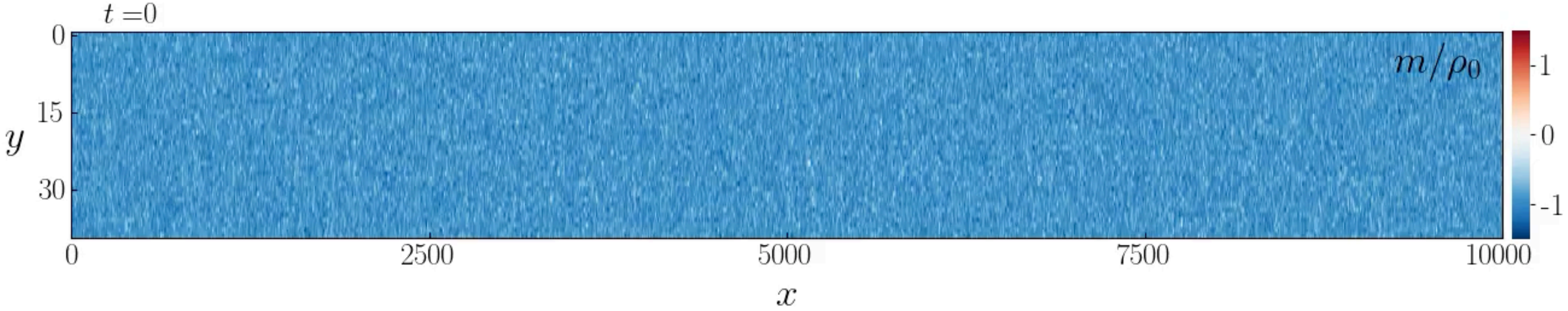




(a)
m/ρ_o
$t = 400$
(b)
$t = 2200$
(c)
ρ/ρ_o
p
(d)
P_r
L
200
400
800
r_c
r
(e)
r_c
T
(f)
D
(g)
τ_d
L
x
y

Spontaneous Transition

In 2D



Analytic Description

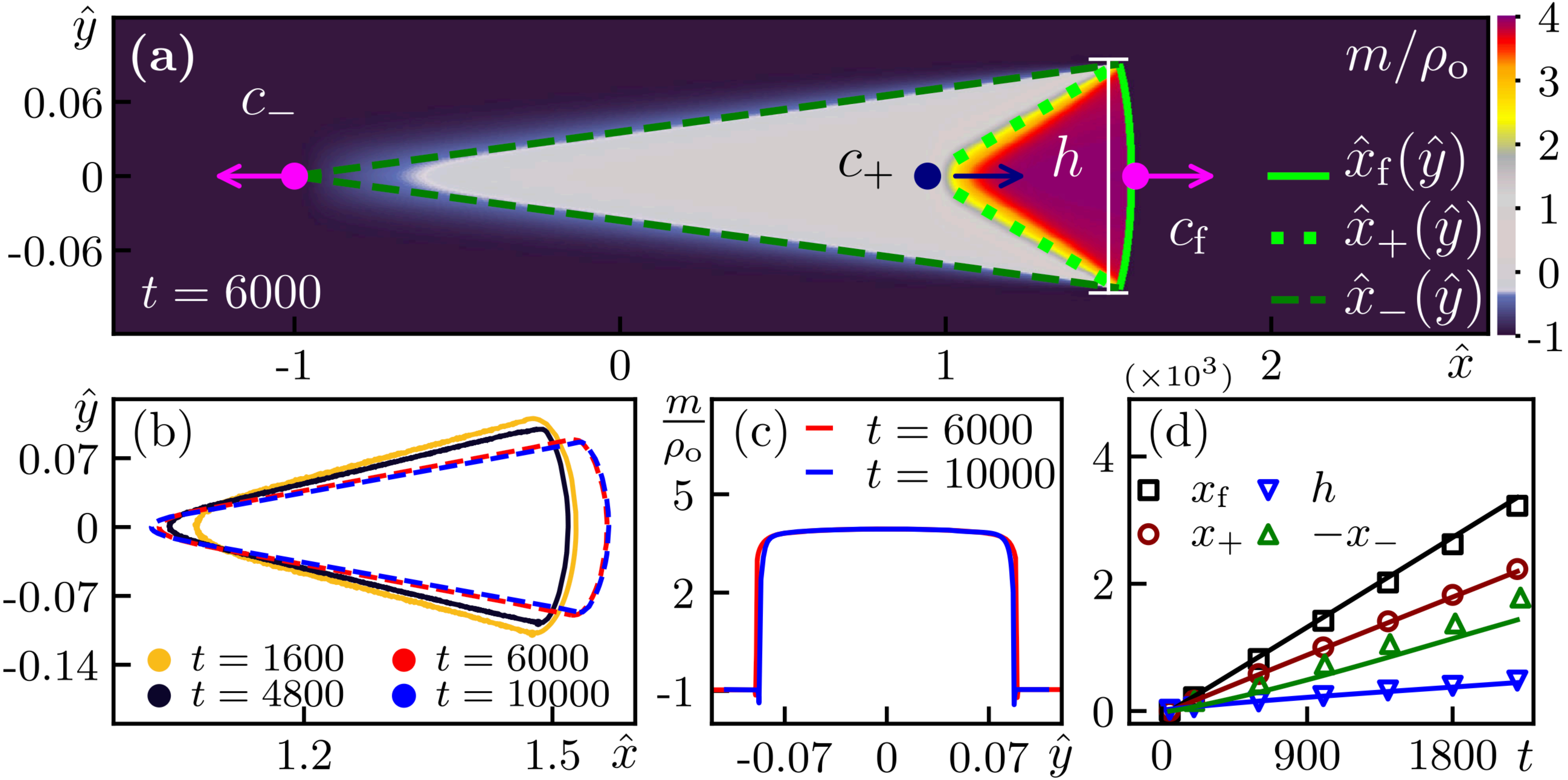
Not in detail

- Hydrodynamic Equations defined for magnetization and density

$$\partial_t \rho = -v\partial_x m + \nabla \cdot \overrightarrow{D}\nabla \rho \qquad \partial_t m = -v\partial_x \rho + \nabla \cdot \overrightarrow{D}\nabla m + F(\rho, m)$$

- Numerical integration gives solutions that fit simulations

- Three domain walls characterized by the above equations


(a)
$\hat{y}$
c_-
c_+
h
c_f
m/ρ_o
$\hat{x}_\mathrm{f}(\hat{y})$
$\hat{x}_+(\hat{y})$
$\hat{x}_-(\hat{y})$
$t = 6000$
0.06
0
-0.06
-1
0
1
2
$\hat{x}$
4
3
2
1
0
-1
(b)
$t = 1600$
$t = 4800$
$t = 6000$
$t = 10000$
0.07
0
-0.07
-0.14
1.2
1.5
(c)
$\frac{m}{\rho_\mathrm{o}}$
$t = 6000$
$t = 10000$
5
2
-1
-0.07
0
0.07
(d)
$(\times 10^3)$
x_f
h
x_+
$-x_-$
4
2
0
0
900
1800
t

Summary Of The Results

- The critical radius
  - Independent of system size
  - Shrinks with decreasing temperature
  - linear with diffusion
- Spontaneous transition possible with low diffusion
  - time to nucleation $\tau \sim L^{-2}$

Conclusion:
Flocking phase is metastable in thermodynamic limit.

Further Research

- More complex geometry & movement
- Large systems
  - Interplay of multiple nucleations
- Continuous symmetries
  - Directional polarization?

Thank you